MIL-C-15370C
AMENDMENT 2
14 June 1976
SUPERSEDING
AMENDMENT 1
29 December 1972

MILITARY SPECIFICATION

COUPLERS, DIRECTIONAL (COAXIAL LINE OR WAVEGUIDE),

GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-C-15370C, dated 7 April 1971, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title, delete and substitute:

"COUPLERS, DIRECTIONAL (COAXIAL LINE, WAVEGUIDE, OR PRINTED CIRCUIT)

GENERAL SPECIFICATION FOR"

1.1, lines 4 and 5, delete and substitute: "levels, the radio-frequency energy transmitted therein (see 6.1)."

PAGE 2

2.1, list of Military Specifications, add:

"MIL-C-55302 - Connectors, Printed Circuit Subassembly and Accessories."

PAGE 3

3.3.4 Delete "temperature range of -65° to +95°C." and substitute: "specified temperature range (see 3.1)."

3.4.2, delete and substitute:

"3.4.2 Connectors. Primary-line and secondary-line connectors shall be as specified (see 3.1). The material and gaging for receptacle connectors shall conform to the requirements of MIL-C-39012, Drawing REA 49330 or RFA 49331, or MIL-C-55302, as applicable."

3.6, line 1: Delete "Coupling." and substitute "Coupling factor."

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3.7, line 2: After "exceed" delete and substitute "the limits specified (see 3.1)."

*Add the following new requirements:

"3.18 Shock. When directional couplers are tested as specified in 4.6.13, the VSWR, effective directivity, and coupling factor shall be as specified (see 3.1).

3.19 Average coupling (applicable to hybrid couplers only). When directional couplers are tested as specified in 4.6.14, the arithmetic average of the measured coupling (in dB) to the two outputs shall be as specified (see 3.1) over the specified frequency range.

3.20 Amplitude equality (applicable to hybrid couplers only). When directional couplers are tested as specified in 4.6.15, the difference between the measured coupling at the two outputs (in dB) shall be as specified (see 3.1) over the specified frequency range. Where the limit is expressed as ± X dB this shall be interpreted to allow a maximum difference of 2X dB.

3.21 Phase difference (applicable to hybrid couplers only). When directional couplers are tested as specified in 4.6.16, the difference in phase between the two outputs shall be as specified (see 3.1) over the specified frequency range.

3.22 Isolation (applicable to hybrid couplers only). When directional couplers are tested as specified in 4.6.17, the ratio of the input power to the power from the isolated port shall be as specified (see 3.1) over the specified frequency range ."

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TABLE I, line 3: Following "coupling" add "factor".

*TABLE I, after "Vibration, high frequency" add:

"Shock	3.18	4.6.13
Average coupling (when specified)	3.19	4.6.14
Amplitude equality (when specified)	3.20	4.6.15
Phase difference (when specified)	3.21	4.6.16
Isolation (when specified)	3.22	4.6.17 "

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TABLE II, line 2: Following "coupling" add "factor".

*TABLE II, after "coupling, variation" add:

"Average coupling (when specified)	3.19	4.6.14	1.0	---
Amplitude equality (when specified)	3.20	4.6.15	1.0	---
Phase difference (when specified)	3.21	4.6.16	1.0	---
Isolation (when specified)	3.22	4.6.17	1.0	--- "

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4.6.3, line 1: Delete "Coupling (see 3.6)" and substitute "Coupling factor (see 3.6)."

PAGE 8

4.6.9, delete and substitute:

"4.6.9 Temperature cycling (see 3.12). Directional couplers shall be tested in accordance with method 102, test condition D, of MIL-STD-202, except temperature range shall be as specified (see 3.1)."

*Add the following new tests:

"4.6.13 Shock(see 3.18). Unless otherwise specified (see 3.1), directional couplers shall be tested as specified in 4.6.13.1. When specified (see 3.1), directional couplers shall be tested as specified in 4.6.13.2.

4.6.13.1 Specified pulse. Directional couplers shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

(a) Mounting - Rigidly mounted by flanges or mounting holes.
(b) Test condition - As specified (see 3.1).
(c) Measurement after test - After the test, the directional couplers shall be subjected to VSWR, effective directivity and coupling factor as specified in 4.6.5, 4.6.2 and 4.6.3, respectively.

4.6.13.2 High impact. Directional couplers shall be tested in accordance with method 207 of MIL-STD-202. The following details shall apply:

(a) Mounting - Rigidly mounted by flanges or mounting holes on fixture of figure 207- 5.
(b) Measurements after test - As specified in 4.6.13.1(c).

4.6.14 Average coupling (see 3.19). The average coupling of directional couplers shall be tested using either or both the swept frequency technique and network analyzer. All ports shall be terminated in a 50 ohm (nominal) load during the test. Any port may be used as the input port.

4.6.15 Amplitude equality (see 3.20). The amplitude equality of directional couplers shall be tested as specified in 4.6.14.

4.6.16 Phase difference (see 3.21). The phase difference of directional couplers shall be tested as specified in 4.6.14.

4.6.17 Isolation (see 3.22). The isolation of directional couplers shall be tested as specified in 4.6.14."

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*NOTE: The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - SL
Navy - OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - WC
Navy - MC, CG, AS, OS, SH
Air Force - None

Preparing activity:
Navy - EC

Agent:
DSA - ES

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